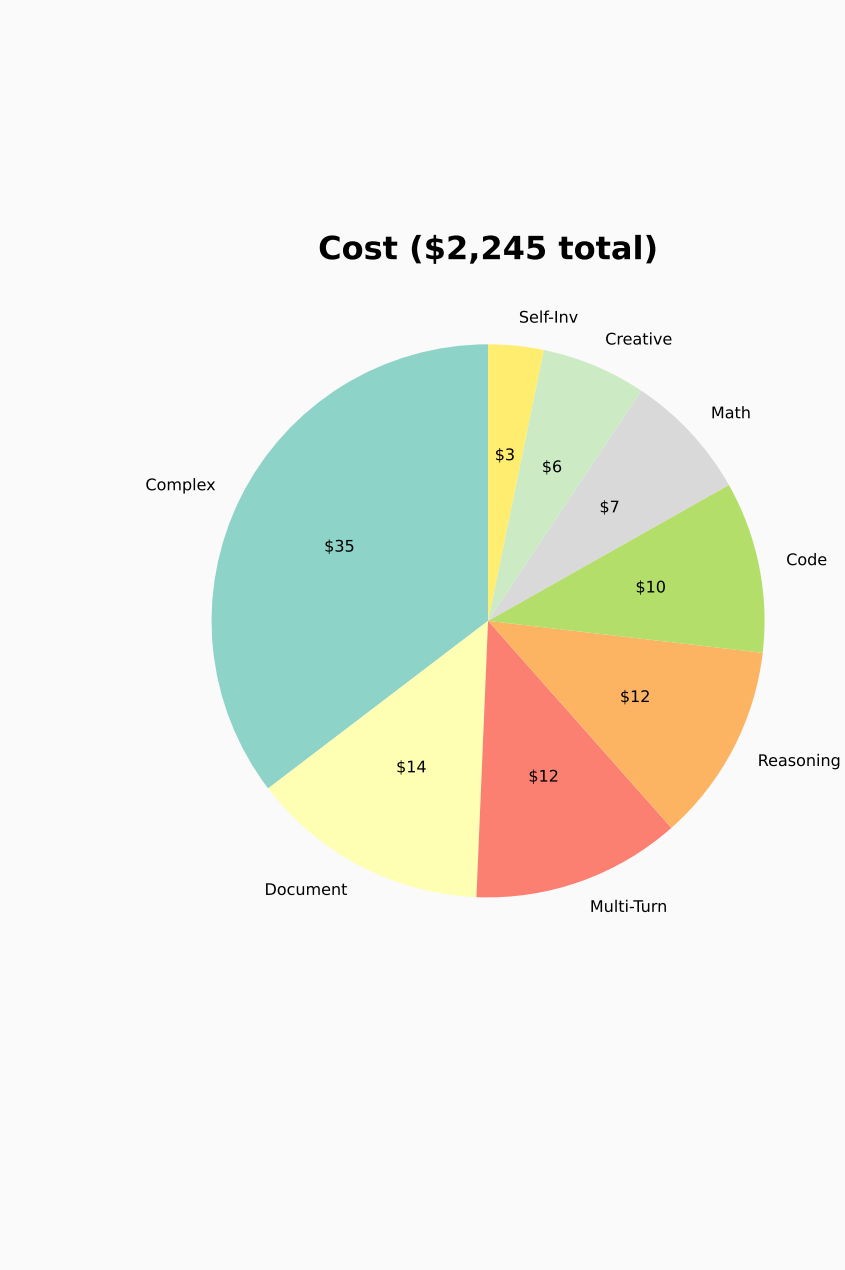
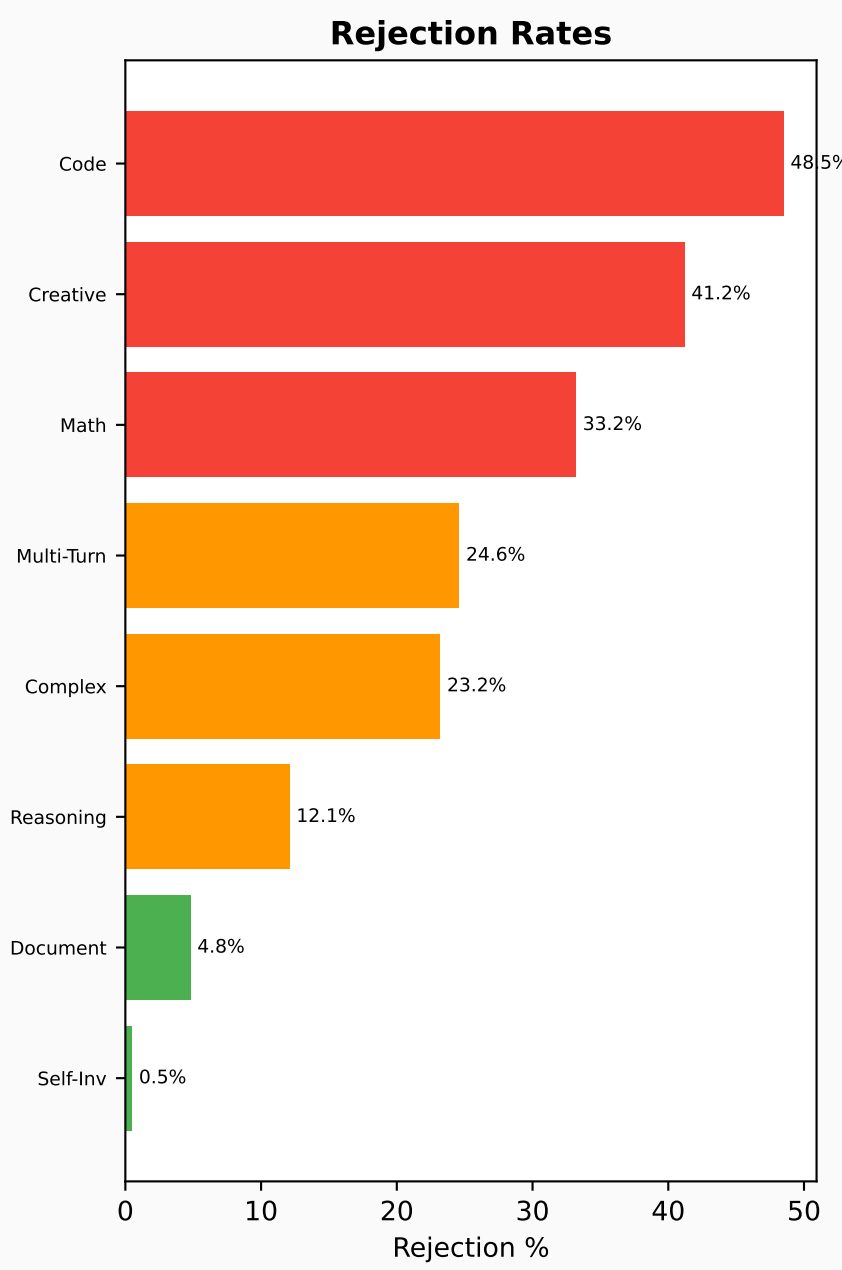
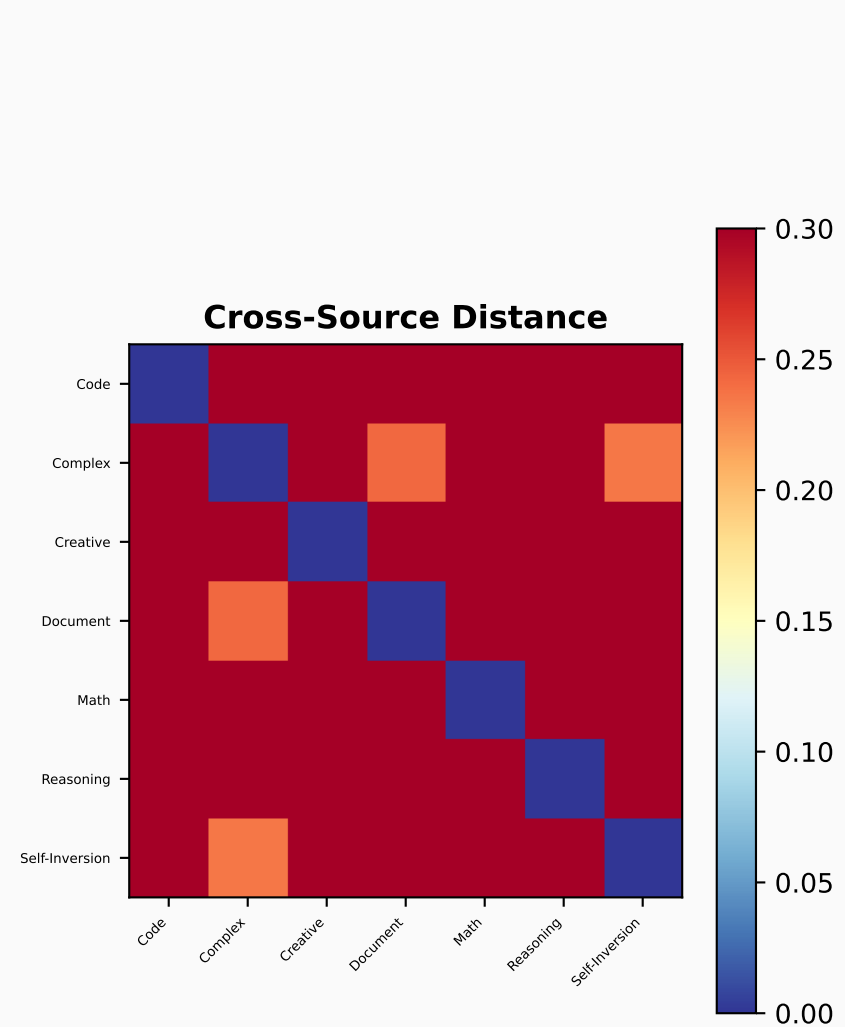
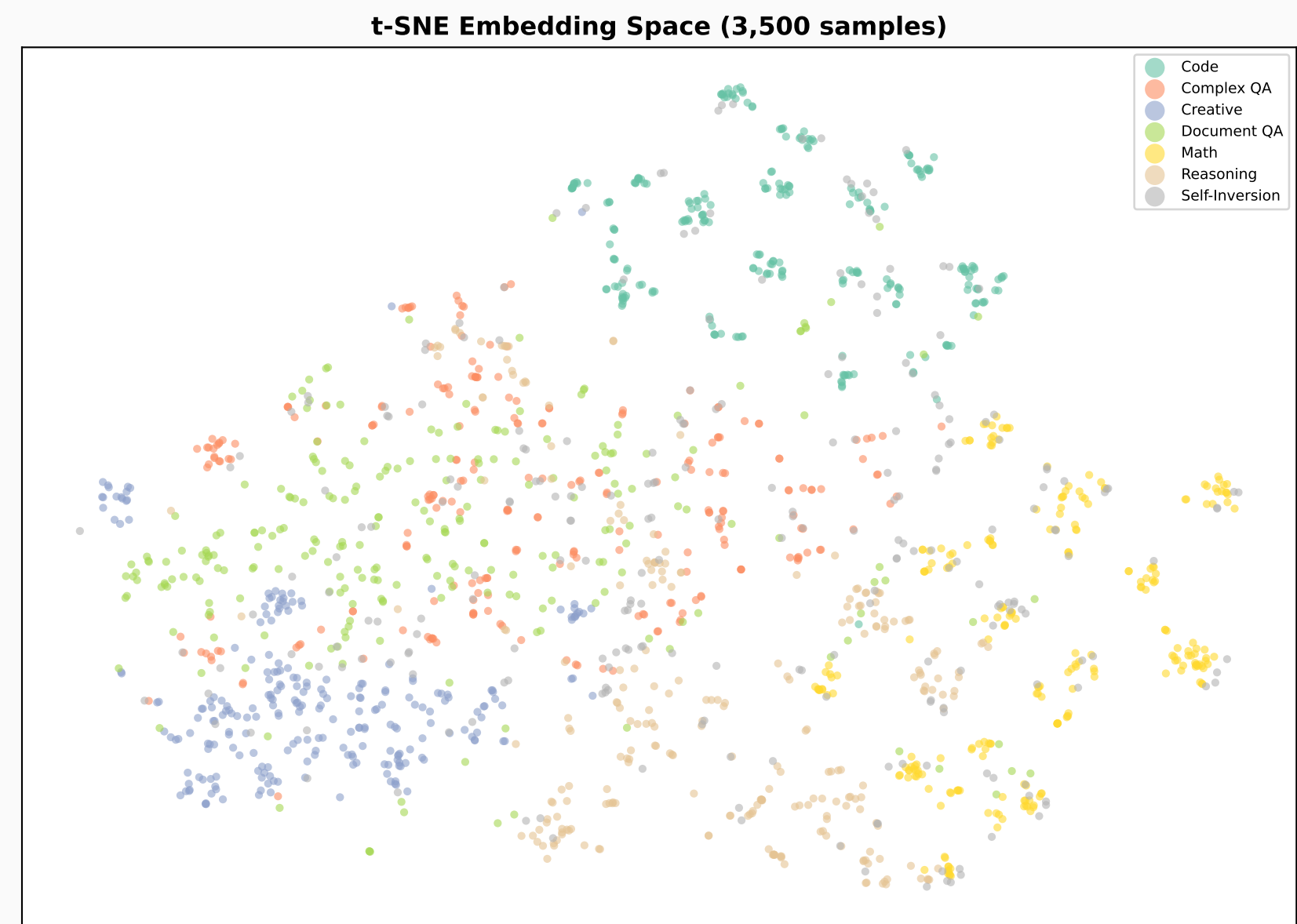
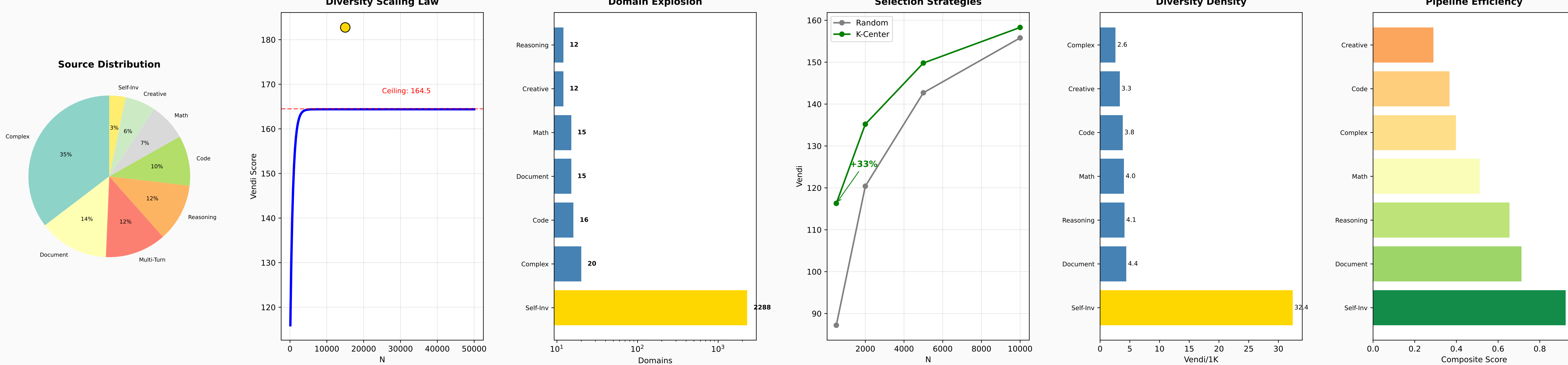


FANNO-Dev: Diversity-Verified Large-Scale Instruction Data Synthesis

153,351 samples · 8 pipelines · Vendi Score 182.75 · 2,297 domains · \$2,245 total cost



Quality Scorecard		
Metric	Value	Grade
JSON Valid	100%	A+
Fields Complete	100%	A+
Refusal	0.01%	A+
Contamination	0	A+
Quality Score	>72	A
TTR Stability	0.84	A
Rejection	27.4%	B+

TOP 10 KEY FINDINGS

● Vendi Score 182.75 — first quantitative diversity measurement for instruction data

● Exponential saturation scaling: $V(N) = 55.4(1 - e^{-N/745}) + 109$, $R^2=0.996$

● Self-Inversion: 2,288 domains (143× explosion), 87.6% hull coverage, in ALL 20 clusters

● Cross-source cosine distance 0.961 — nearly orthogonal pipeline subspaces

● K-Center-Greedy: +33% diversity at N=500; random near-optimal at scale

● Self-Inv diversity density: 32.4 Vendi/1K (12× more than Complex QA)

● Document QA: #1 linguistic diversity; Code QA: most semantically unique

● Tag space utilization only 4.1% — room for 24× growth through template expansion

● 0 benchmark contamination verified across 26,739 test instances

● Total cost: 2,245(15.58/1K clean samples) using GPT-4o (17 Azure endpoints)

DELIVERABLES

Deliverable	Details
135+ Figures	Publication-quality PNG+PDF pairs
27 LaTeX Tables	Paper-ready formatting with captions
57 JSON Reports	Machine-readable analysis results
LaTeX Paper Draft	Complete 8-section paper with refs
Alpaca Format	125,280 samples for standard training
ShareGPT Format	141,901 samples with multi-turn
Training Recipe	LLaMA-Factory compatible configs
5 Selection Strategies	From random to K-Center-Greedy
Contamination Verified	5 benchmarks, 0 matches
130+ Git Commits	Full history on dev branch

EFFICIENCY RANKING

Rank	Pipeline	Density
#1	Self-Inversion	32.4 V/1K
#2	Document QA	4.4 V/1K
#3	Reasoning QA	4.1 V/1K
#4	Math QA	4.0 V/1K
#5	Code QA	3.8 V/1K
#6	Creative	3.3 V/1K
#7	Complex QA	2.6 V/1K